

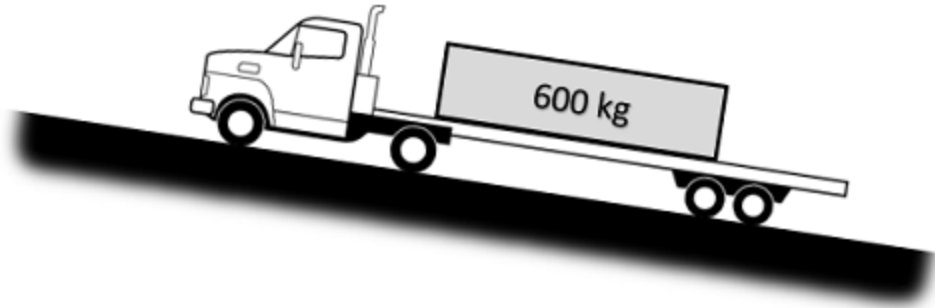
Lesson 6: Newton's 2nd Law

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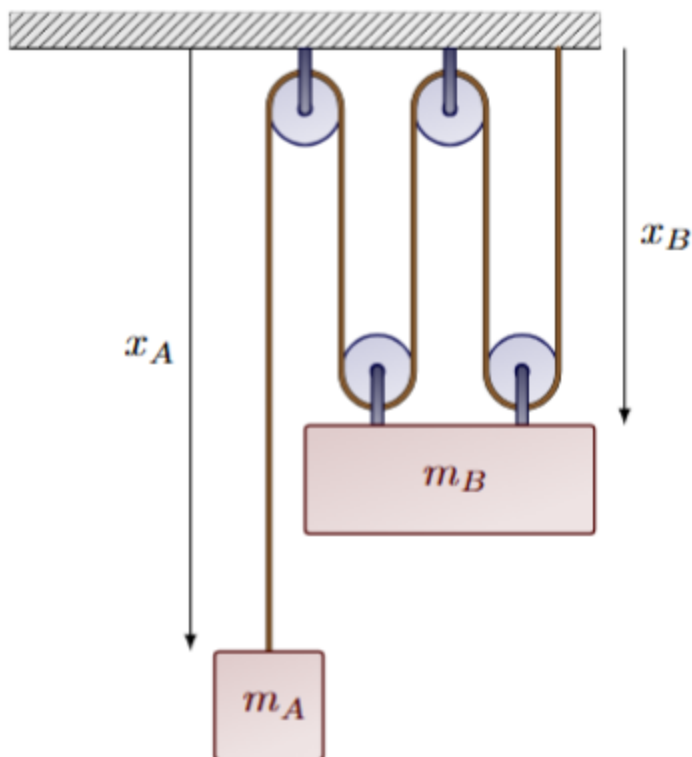
1. A 50kg box is being pulled across a surface by a 200 N force in the direction shown below. If the static and kinetic coefficient of friction is assumed to be 0.3, what is the rate of acceleration of the box? Assuming the box starts at rest how far will the box move in a three second period? /1



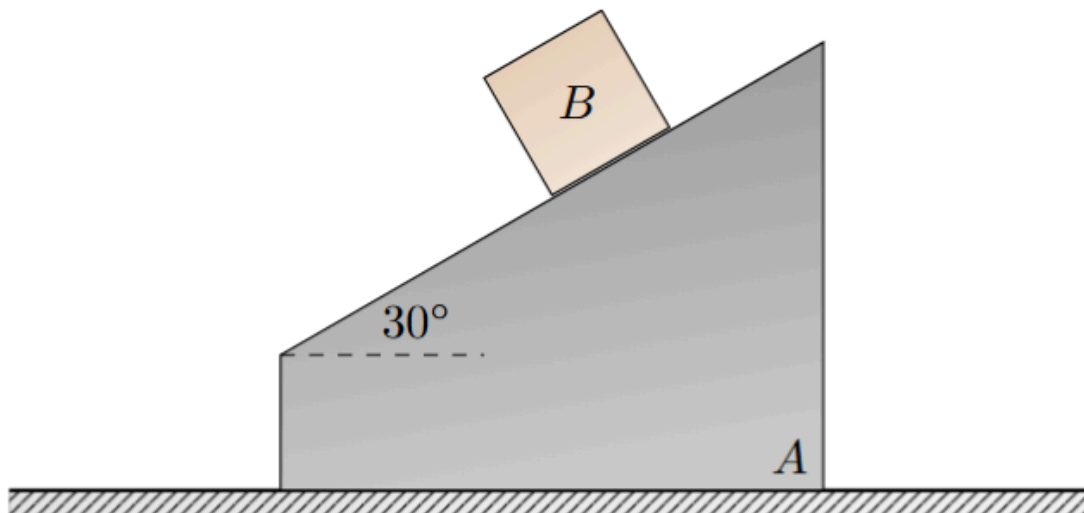
2. A man in a flatbed truck starts from rest up a hill at an angle of 10° degrees. If he is carrying a 600 kg crate in the back and the static coefficient of friction is .3, what is the rate of acceleration at which we would expect the crate to begin sliding off of the back of the truck? How long will it take the truck to reach a speed of 25 m/s at this rate? /1



3. In the pulley system shown below mass A is 10 kg and mass B is 12 kg. Find the acceleration, in m/s^2 , of each block and the tension, in N, in the cable. When reporting the acceleration, use a minus sign if a block is accelerating downward. /1



4. The 18-lb block B slides on the 50-lb wedge A . The block B and wedge A both start from rest. Friction can be neglected between all surfaces. Find the acceleration of wedge A . /1



Answer: _____